Five-variable weak Markus–Yamabe: graded compression and affine-Hessian obstructions

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Abstract

We investigate the construction of a polynomial vector field $X : \mathbb{R}^5 \to \mathbb{R}^5$ whose Jacobian has characteristic polynomial $(\lambda+1)^5$ everywhere and which has two distinct real zeros. We do not obtain such a field. We prove that two complete bigraded compression families built from a known six-dimensional example contain no field with the requested characteristic polynomial. Exact polynomial-identity certificates establish the exclusions. In the final chain proof, eight affine-linear coefficient equations force the variables and a ninth has exact nonzero residual. We also prove a uniform Hessian-span theorem: for a specified linear form A in three auxiliary variables, every potential $\kappa A^2 + D$, with $\kappa \neq 0$ and arbitrary D affine in those variables, resists every invertible affine source–target normalisation to constant Jacobian trace. A separate checker reconstructs the Jacobians and replays the finite certificates. A tempting factor-sharing compression retains three zeros but has an explicit positive Jacobian eigenvalue. None of these conclusions rules out a five-dimensional counterexample outside the stated architectures.

1 Target, source boundary and status

The target is an explicit real polynomial map satisfying

$$\det(\lambda I_5 - DX(z)) = (\lambda+1)^5 \quad (z \in \mathbb{R}^5), \qquad X(z_1) = X(z_2) = 0, \quad z_1 \neq z_2. \tag{1}$$

Writing $X = -\operatorname{Id} + H + \eta$, with constant η, makes the spectral requirement

$$\det(LI_5 - DH) = L^5. \tag{2}$$

The constant η affects fibers but not the Jacobian. Every exclusion below concerns (2); it does not exclude a Hurwitz field with nonconstant characteristic polynomial.

The source audit, updated on 19 October 2026, includes the fourteen-dimensional chain construction of Castañeda, Honorato and Valenzuela-Henríquez [2], the six-dimensional construction posted by liqsweep [1], the five-variable Hessian potential of Meng and Yang [3], a degree-bounded retargeting obstruction [4], and Liu's Hesse-system formulation [5]. Earlier work already discusses failed chain shortening and coefficient-span methods. The contributions here are the exact displayed family exclusions, their transparent certificates, and the uniform affine-trace obstruction below. This is a targeted source boundary, not an exhaustive priority audit. The six-dimensional identities and its three listed zeros were reproduced independently. The new results have not undergone external specialist review or proof-assistant formalisation.

Result status. The requested five-dimensional construction remains unresolved. The deliverables are exact obstruction theorems for specified architectures, a rejected candidate, and reproducible exact evidence. Broader uncertified searches are reported only as supplementary exploration.

2 The baseline and the failed direct substitution

For base variables (x, y, z) put

$$q = y - 6xz, \qquad f = x^2(3q + 2xz), \tag{3}$$

$$g_2 = 24xq(q + xz), \qquad g_3 = 2q(2q + 3xz), \tag{4}$$

$$g_4 = 12q^2 f, \qquad g_6 = 4q^3 f.$$

The remaining base polynomial is

$$g_5 = 2xq^2(7q + 6xz).$$

The six-dimensional field in [1] is

$$X_6 = \begin{pmatrix} -x + f \\ 1 - y + a - g_2 \\ -z + b - g_3 \\ -a - g_4 \\ -b + c - g_5 \\ -c - g_6 \end{pmatrix}. \tag{5}$$

Its characteristic polynomial is $(\lambda + 1)^6$. The checker independently expands this identity and verifies the zeros

$$(0, 1, -4, 0, 0, 0), \quad (-1, -11, 2, 12, -6, 4), \quad (1, 7, 5/4, -3, -3/2, 1/2).$$

The proof mechanism in the source combines a constant determinant with a weighted similarity identity. Its auxiliary variables have weights $2, 2, 3$ under the base weights $(0, 1, 1)$.

The factorisation $g_6 = (q/3)g_4$ suggests sharing an auxiliary variable. After scaling the old a by three, the tempting five-dimensional field is

$$X_{\mathrm{try}} = \begin{pmatrix} -x + f \\ 1 - y + 3a - g_2 \\ -z + b - g_3 \\ -a - 4q^2 f \\ -b + qa - g_5 \end{pmatrix}. \tag{6}$$

It retains the three zeros

$$(0, 1, -4, 0, 0), \quad (-1, -11, 2, 4, -6), \quad (1, 7, 5/4, -1, -3/2).$$

It is nevertheless not a Hurwitz field.

Proposition 2.1. *At $P = (1, 0, 0, -1, 0)$ the field (6) has a positive Jacobian eigenvalue. More precisely,*

$$\det(\lambda I - DX_{\mathrm{try}}(P)) = (\lambda + 1)^3\big((\lambda + 1)^2 - 6\big).$$

Proof. Direct differentiation gives

$$DX_{\mathrm{try}}(P) = \begin{pmatrix} -1 & 3 & -16 & 0 & 0 \\ 0 & -1 & 0 & 3 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 6 & 0 & -1 \end{pmatrix}.$$

Expansion gives the stated polynomial. The eigenvalue $-1 + \sqrt{6}$ is positive. □

This failure identifies an important distinction: eliminating auxiliary variables in the equilibrium equations can preserve a fiber without preserving the Jacobian away from that fiber. A determinant argument that evaluates an auxiliary variable at its equilibrium value does not establish a global Jacobian identity.

3 Architectural coverage and relation to earlier obstructions

The next two families are not arbitrary degree-bounded searches. They are the complete polynomial families with the displayed pairs of component gradings, after fixing the inherited base terms and two nonzero auxiliary feedthroughs. The feedthrough coefficients can be normalised to one by diagonal rescaling of the auxiliary coordinates; this is a linear conjugacy and hence preserves the Jacobian characteristic polynomial. Cases in which either feedthrough is zero are not covered by that normalisation.

Proposition 3.1. *For the parallel grading, the five components admit respectively* $2, 5, 4, 8, 8$ *monomials. Fixing the eight inherited base coefficients and two feedthrough coefficients leaves* $27 - 8 - 2 = 17$ *free coefficients. For the chain grading, the corresponding counts are* $2, 4, 4, 7, 11$*; fixing the same data and imposing the two trace relations leaves* $28 - 8 - 2 - 2 = 16$ *free coefficients. Thus the formulae in Sections 4 and 5 exhaust their stated bigraded architectures.*

Proof. All variables except x have positive first weight, while x has second degree one. For each component, enumerate the finitely many exponents of the positive-weight variables and solve the second-degree equation for the exponent of x. Retaining the nonnegative solutions gives the displayed lists and counts. The exact inventories are reproduced by `discovery/grading_audit.py`. □

Castañeda, Honorato and Valenzuela-Henríquez already describe failed shortening of their ordinary chain presentation [2]. A later degree-filtration result excludes a different retargeting strategy only for target polynomials of degree at most eight [4]. Neither statement contains the two bigraded families below. Conversely, Proposition 3.1 does not show that a successful five-dimensional construction must respect either grading, retain the three fixed base components, or use these auxiliary architectures. For example, an a^2 term in the fourth parallel component has first weight 4 rather than the required component weight 3 and is therefore outside that family; so are changes to the fixed polynomial f and branches with a zero feedthrough. The exclusions remove two natural complete search spaces, not the global problem.

4 Parallel auxiliaries: an exact exclusion

Consider the following family, with all coefficients arbitrary in a characteristic-zero field:

$$H = \begin{pmatrix} f \\ a + \tau x b - g_2 \\ b - g_3 \\ -P + Aa + Bb \\ -Q + Ca + Db \end{pmatrix}, \tag{7}$$

where

$$P = \sum_{k=0}^{3} p_k x^{k+2} y^{3-k} z^k, \qquad Q = \sum_{k=0}^{3} r_k x^{k+1} y^{3-k} z^k, \tag{8}$$

$$A = \alpha xy + \beta x^2 z, \qquad B = \gamma x^2 y + \delta x^3 z, \tag{9}$$

$$C = \epsilon y + \zeta xz, \qquad D = \eta_1 xy + \eta_2 x^2 z. \tag{10}$$

This is a 17-parameter family: eight cubic coefficients, eight auxiliary-coupling coefficients and τ. The first three nonlinear base terms are fixed to the baseline. The family allows a cross-coupling τxb in the second component.

Every displayed term has componentwise weighted degree $w_i + 1$ for $w = (0, 1, 1, 2, 2)$ and degree ν_i for $\nu = (1, -1, -2, -1, -2)$. These two gradings explain the finite monomial lists; they are explicit restrictions, not properties of all polynomial fields.

Theorem 4.1. *No choice of the 17 coefficients in* (7) *satisfies* $\det(LI - DH) = L^5$. *Consequently no constant translation of* $-\mathrm{Id} + H$ *supplies* (1) *in this family.*

Proof. The fixed base terms satisfy

$$f_x - (g_2)_y - (g_3)_z = 0.$$

Vanishing trace therefore forces $D = -A$, so $\eta_1 = -\alpha$ and $\eta_2 = -\beta$. The coefficients linear in a, b in the next characteristic coefficient force

$$\zeta = -\alpha - \tau\epsilon, \qquad \gamma = \beta + \tau\alpha - 3\tau. \tag{11}$$

The checker reconstructs the unreduced family, checks its trace and the two relevant coefficients of the second elementary symmetric function, and verifies that these substitutions yield exactly the model used in the certificates. Thus it suffices to examine the resulting 13-parameter family.

Differentiate *before* setting $x = 1$. Expand

$$\det(LI - DH)|_{x=1} - L^5$$

in L, y, z, a, b. There are 45 distinct nonzero coefficient equations in the parameters. Thirteen do not involve $p_0, \ldots, p_3, r_0, \ldots, r_3$. Denote these thirteen polynomials by $E_1, \ldots, E_{13}$.

The rational identity certificate `cross_pure_certificate.json` proves

$$\beta^2,\ \epsilon^2\ \in (E_1, \ldots, E_{13}). \tag{12}$$

It supplies 21 intermediate polynomial identities. The verifier reconstructs the E_i from the displayed Jacobian; it does not accept a separate list of unverified equations as the input to the proof.

At a simultaneous zero of the coefficient equations, (12) implies $\beta = \epsilon = 0$. One of the thirteen equations is

$$-\alpha^2 - \alpha\epsilon\tau - \beta\epsilon = 0,$$

so $\alpha = 0$. A further coefficient, after these substitutions, is

$$8\delta - 128\tau = 0.$$

Every possible solution must consequently obey

$$\alpha = \beta = \epsilon = 0, \qquad \delta = 16\tau. \tag{13}$$

The remaining auxiliary coupling is then $B = -\tau f$, and $A = C = D = 0$.

Substitute (13) into the 45 coefficient polynomials and discard zeros and duplicates. The resulting 18 polynomials $F_1, \ldots, F_{18}$, in the nine variables $p_0, \ldots, p_3, r_0, \ldots, r_3, \tau$, satisfy

$$1 \in (F_1, \ldots, F_{18}). \tag{14}$$

The certificate `cross_final_certificate.json` proves (14) through 31 further rational polynomial identities. A common zero would turn (14) into $1 = 0$, a contradiction.

Specialising $x = 1$ loses no force in this exclusion: a global characteristic identity would necessarily hold on that slice. No claim that checking one slice establishes a global identity is used. □

Remark 4.2. The theorem excludes arbitrary values of the displayed coefficients, not merely integer coefficients within a searched box. It does not exclude terms of other bidegrees, changes to the fixed base terms, different dependence on the auxiliary variables, or five-dimensional constructions unrelated to this baseline.

5 An unequal-weight chain: a second exclusion

A different attempt shares the first auxiliary between the second and third components, with auxiliary weights $2, 3$. Let

$$P = \sum_{k=0}^{3} p_k x^{k+1} y^{3-k} z^k, \qquad Q = \sum_{k=0}^{4} v_k x^{k+2} y^{4-k} z^k, \tag{15}$$

$$S = \sum_{k=0}^{2} w_k x^{k+2} y^{2-k} z^k, \qquad R = rxy + sx^2 z. \tag{16}$$

After imposing the trace condition, the family is

$$H = \begin{pmatrix} f \\ \tau x a - g_2 \\ a - g_3 \\ b + P + Ra \\ Q + Sa + jx^2a^2 - Rb \end{pmatrix}. \tag{17}$$

Its sixteen coefficients are unrestricted. The weights are $(0, 1, 1, 2, 3)$ and the second grading is $(1, -1, -2, -2, -2)$. The quartic terms and the quadratic term jx^2a^2 are included; simply omitting them would give a substantially narrower test.

Theorem 5.1. *No member of* (17) *satisfies* $\det(LI - DH) = L^5$.

Proof. Differentiate first and specialise $x = 1$. Among the 38 distinct nonzero characteristic coefficient polynomials are

$$\begin{aligned} E_0 &= -2j - r\tau - s - 3\tau, \\ E_1 &= r\tau + s, \\ E_2 &= -45r\tau + 248r - 8s\tau + 42s, \\ E_3 &= 32j + 16r\tau + 3s\tau, \\ E_4 &= -\tau(16r + 3s). \end{aligned}$$

Their vanishing gives $32j = E_3 + E_4 = 0$. Next $E_0 + E_1 = -2j - 3\tau$ gives $\tau = 0$, then E_1 gives $s = 0$, and finally E_2 gives $r = 0$.

After substituting $j = \tau = s = r = 0$, 27 distinct nonzero coefficient equations remain in the twelve coefficients of P, Q, S. All are affine-linear. Eight of them successively give

$$w_0 = w_1 = w_2 = 0, \qquad p_1 = 132, \quad p_2 = -744, \quad p_3 = 1392, \qquad v_3 = -1920, \quad v_4 = 3264.$$

A further necessary equation is

$$-93312w_2 + 25344v_4 + 34560v_3 = 0.$$

The forced values make its left-hand side

$$25344(3264) + 34560(-1920) = 16\,367\,616 \neq 0,$$

which is the required contradiction. The exact equation indices and a checked rational linear combination are recorded by `check_chain_short_identity.py`. The fuller thirteen-step unit-ideal certificate is retained as redundant machine-checkable evidence. □

6 A uniform affine-Hessian obstruction

For variables (x, y, a, b, c) set $u = 1 + xy$ and, for nonzero r, t, define

$$A_{r,t} = au^3 + rxbu^2 - tx^3c.$$

Let $d_0, d_1, d_2, d_3 \in \mathbb{R}[x, y]$ be arbitrary and put

$$\Phi = \kappa A_{r,t}^2 + d_0(x, y) + ad_1(x, y) + bd_2(x, y) + cd_3(x, y), \qquad \kappa rt \neq 0. \tag{18}$$

The coefficient-span object used below is the Hesse system in the terminology of Liu [5]; that general linear-algebra criterion is not claimed as new. The contribution is the following full-span calculation and its uniformity over the perturbations in (18).

Theorem 6.1. *If a constant real 5×5 matrix M satisfies*

$$\operatorname{tr}\big(M \operatorname{Hess} \Phi(w)\big) = \textit{constant} \quad (w \in \mathbb{R}^5),$$

then $M + M^{\mathsf{T}} = 0$, and the constant is zero. Equivalently, the nonconstant Hessian coefficient matrices of Φ span $\operatorname{Sym}_5(\mathbb{R})$.

Proof. Only the symmetric part $N = (M + M^{\mathsf{T}})/2$ contributes to the trace. Split N into the symmetric block on (x, y), the mixed block between (x, y) and (a, b, c), and the symmetric block on (a, b, c). Compare coefficients in descending auxiliary degree.

The coefficients of b^2, a^2, x^2a^2 first give

$$2\kappa r^2 n_{xx} = 0, \qquad 12\kappa n_{xy} = 0, \qquad 30\kappa n_{yy} = 0,$$

so the base block vanishes. Terms of auxiliary degree one then give

$$4\kappa r n_{xa} = 4\kappa r n_{xb} = 24\kappa n_{ya} = -12\kappa t n_{yc} = 0$$

and the two equations

$$-12t n_{xc} + 20r n_{yb} = 0, \qquad -16rt n_{xc} + 16r^2 n_{yb} = 0.$$

Their coefficient determinant is $128r^2t \neq 0$, so the mixed block vanishes. Finally the coefficients of $xy, x, x^3, x^2, x^4, x^6$ give, respectively,

$$12\kappa n_{aa}, \quad 4\kappa r n_{ab}, \quad -4\kappa t n_{ac}, \quad 2\kappa r^2 n_{bb}, \quad -4\kappa rt n_{bc}, \quad 2\kappa t^2 n_{cc},$$

and hence the last block vanishes.

The perturbation in (18) cannot alter this triangular argument. Its second base derivatives have auxiliary degree at most one, its mixed derivatives have auxiliary degree zero, and its second auxiliary derivatives vanish. At each stage its possible contributions therefore lie in blocks already eliminated or below the active diagonal block. Thus $N = 0$. Conversely, every skew-symmetric matrix pairs to zero with every symmetric Hessian. The constant is therefore zero.

As an exact check, the selected 15×15 coefficient minor has determinant

$$-1\,304\,596\,316\,160\ \kappa^{15} r^{10} t^6, \tag{19}$$

which is nonzero under the hypotheses. At $(\kappa, r, t) = (1, 3, 1)$ its absolute value is

$$77\,035\,107\,872\,931\,840.$$

□

Corollary 6.2. *For every $S, T \in \mathrm{GL}_5(\mathbb{R})$ and $c_0, d_0' \in \mathbb{R}^5$, the map*

$$X(z) = T\nabla\Phi(Sz + c_0) + d_0'$$

has nonconstant Jacobian trace. In particular it cannot satisfy (1).

Proof. Cyclic invariance gives

$$\operatorname{tr} DX(z) = \operatorname{tr}\big(ST \operatorname{Hess}\Phi(Sz + c_0)\big).$$

Surjectivity of the source affine transformation and Theorem 6.1 would force the invertible matrix ST to be skew-symmetric. Every skew-symmetric matrix of odd order is singular, a contradiction. □

The audited version of the Meng–Yang construction [3] uses $A = A_{3,1}$ and a potential whose remaining terms are affine in (a, b, c); every nonzero coefficient of A^2 is covered by (18). This formulation deliberately avoids relying on the numerical coefficient of the source's linear-in-A term. More generally, nonzero r, t are coordinate rescalings of $r = 3, t = 1$, so this part records coordinate robustness rather than a separate novelty claim. The arbitrary polynomials d_i are the substantive uniform extension. They need not preserve a constant Hessian determinant or a gradient collision. The theorem is only an affine constant-trace obstruction; it does not exclude nonlinear transformations, other leading quadratic forms or general five-dimensional Hurwitz maps.

7 Certificate design and executed checks

A certificate is a finite sequence of identities

$$P_k = \sum_{i<k} c_{ki}\, m_{ki} P_i,$$

where the c_{ki} are exact rational numbers and the m_{ki} are monomials. The initial P_i are coefficients reconstructed from the stated Jacobian. The final targets are β^2, ϵ^2 or 1, as appropriate. The checker only expands these identities. It need not implement, trust or verify a Gröbner-basis search algorithm.

The discovery implementation uses SymPy's characteristic polynomial and a certificate-producing Buchberger calculation. The independently written standard-library checker uses sparse integer/rational polynomial arithmetic and recursive determinant expansion. The checked bridges are therefore:

$$\text{explicit map} \longrightarrow \text{differentiated matrix} \longrightarrow \text{coefficient equations} \longrightarrow \text{rational identities}.$$

No sampled Jacobian evaluations are substituted for a polynomial identity. The lone $x = 1$ restriction is used only as a necessary condition in an impossibility proof.

Obligation	Executed result
Six-dimensional characteristic identity	Zero residual polynomial
Three baseline zeros	Exact substitution passed
Naive compression	Positive-eigenvalue witness verified
Parallel compression family	Both staged certificates passed
Unequal-weight chain	Unit-ideal certificate passed
Short chain contradiction	Eight forcing equations plus a ninth residual equation; 16 367 616
Original Hessian trace matrix	Submitted integer minor reproduced
Uniform Hessian theorem	Symbolic determinant (19) verified
Grading completeness	Both monomial inventories reproduced
Four deliberate certificate corruptions	All rejected

The four mutation tests alter, respectively, a source equation, an intermediate identity, the asserted final target, and the final derivation. All are rejected. These tests detect several failure modes; they do not constitute a formal verification of the checker itself.

To replay the proof-critical checks after extracting the bundle:

```
python3 -I verify.py --report /tmp/wmy5-verification.json
python3 -I test_mutations.py
python3 -I check_chain_short_identity.py
python3 discovery/grading_audit.py
python3 discovery/generalized_hessian_check.py
```

The original verifier, mutation tests and short-chain check require only the Python standard library. The two discovery audits require SymPy. Expected terminal messages are `ALL EXACT CHECKS PASSED` and `ALL CORRUPTIONS REJECTED`. Optional SymPy discovery scripts can regenerate the rational certificates; they are not needed to verify them. These scripts were also executed in a separate copy of the bundle, and the standalone verifier accepted all three regenerated certificates.

The report path above is outside the extracted package so that replay does not overwrite the manifest-bound historical report.

Supplemental broader searches

Exact Wolfram Language calculations returned the unit ideal for several larger systems that allow changes to the lower base coefficients, as recorded in `supplementary/CAS_RESULTS.md`. The source script reconstructs those systems. Small independent rational certificates for those broader systems are *not* included, and the standalone checker does not claim to verify them. Some local Gröbner calculations exceeded an execution cap; those interruptions supply no mathematical conclusion. These two statuses are kept separate.

8 What remains open in this attempt

The global construction problem (1) is not settled. The precise conclusion is that a candidate cannot be obtained by selecting different coefficients within the two displayed compression families.

The entire potential family (18) also cannot be put into the requested constant-spectrum form by invertible affine changes.

A construction outside these families could change the grading, break the second homogeneity, use different nonlinear dependence on auxiliary variables, or start from another Keller map. This report supplies no existence or nonexistence claim for those possibilities. In particular it does not establish that dimension six is minimal for polynomial constant-spectrum Hurwitz counterexamples, or that all five-dimensional Hurwitz maps are injective.

Sanity checks

The rejected candidate shows that preserving all three exhibited zeros is insufficient: its Jacobian has an explicitly positive eigenvalue elsewhere. For the Hessian obstruction the necessary target trace is -5, whereas every constant trace obtainable by a constant left multiplier is zero. For the two compression exclusions the final contradictions are exact ideal membership and the explicit nonzero integer 16 367 616, not numerical residuals or failures to find coefficients.

References

[1] blueberryvertigo [liqsweep]. (2026, September 6). *An integer polynomial Hurwitz counterexample in dimension six* [Manuscript and verification code]. GitHub. https://github.com/blueberryvertigo/ArXiv-tryouts. README blob inspected: 002e811eeba67d7cade25d2d8f3003b9c5c38002.

[2] Castañeda, Á., Honorato, G., & Valenzuela-Henríquez, F. (2026, August 5). *The weak Markus–Yamabe conjecture fails in dimension 14* [Preprint]. arXiv. https://arxiv.org/abs/2608.05392.

[3] Meng, G., & Yang, L. (2026, July 27). *A five-variable counterexample to the Hessian conjecture, and the low-dimensional status of the Jacobian and Hessian conjectures* [Preprint, version 2]. arXiv. https://arxiv.org/abs/2607.22198v2.

[4] Korea Superintelligence Labs. (2026, August 30). *The degree filtration of the Alpöge subalgebra: Retargeting cannot lower the Markus–Yamabe dimension below fourteen* [Preprint, version 1]. Ideosphere. https://ideosphere.ai/papers/markus-yamabe-dim-v1.pdf.

[5] Liu, H. (2026). *On the Hessian conjecture in Lorentzian signature: Constant pivots and Hesse systems* [Preprint]. arXiv. https://arxiv.org/abs/2608.19112.